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A Comprehensive Performance Analysis of a Hyperledger Fabric-powered Blockchain Network for Cross-Border Fund Transfers

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Abstract

Ample acceptance of Blockchain Technology in today's business world has increased number of use cases tremendously beyond cryptocurrencies for which it was originally developed. Hyperledger Fabric, as an enterprise-grade permissioned Blockchain platform, offers a secure and scalable infrastructure for diverse applications, including financial services. Since performance evaluation of the platform play major role in acceptability of the technology for various applications, in this research article we are presenting a comprehensive performance analysis of the Hyperledger Fabric Blockchain platform with the help of standard benchmarking tool-Hyperledger Caliper for Cross-Border Fund Transfer (CBFT) application and results are validated using existing queuing theory based theoretical model for Hyperledger Fabric. The primary objective of this study is to evaluate the most popular Blockchain platform's performance under various workloads and network conditions. By capturing the interactions between network participants and consensus mechanisms, we have presented latency analysis along with the insights into the platform's scalability and efficiency. This research contributes to a deeper understanding of Blockchain technology and aids in the design of more efficient and robust Blockchain-based applications.

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Keywords: Blockchain; Hyperledger Explorer; Hyperledger Fabric; Latency; Throughput

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1. Introduction

The most prominent cryptocurrency-based peer to peer cash system also known as Bitcoin is proposed by Satoshi Nakamoto in 2009 [1]. The real potential of Bitcoin arises from underneath technology worn by it, acknowledged as Blockchain. Blockchain was primarily invented to rise above the challenges imposed due to centralized systems and multiple intermediaries. It maintains the record of transactions on all the peers of the network and updates it continuously when new transactions get executed. Thus, data is updated and stored on every peer hence eliminating the possibility of unauthorized alteration or modification of data. Starting from the very first transaction, certain numbers of transactions are grouped into a block as per predefined block size (1MB in case of Bitcoin) and block is broadcasted to each and every peer in the network when its limit is reached. After one block is formed the next set of transaction is grouped together to form subsequent block and all these blocks are chained together by cryptographic hash; so, the technology is named as Blockchain.

1.1. Types of Blockchain

Blockchain networks can be divided into Cryptocurrency and Enterprise Networks based on their utilization in business. Cryptocurrency Networks use their own cryptocurrencies explicit to just their network. These are virtual currencies and can be used only on digital networks. Major players in this domain are Bitcoin and Ethereum. They are used to process the transactions without any intermediaries [1]. Enterprise Networks are used by enterprises for business purposes [2]. Examples of these networks are Hyperledger Fabric, R3 Corda etc.

1.2 Purpose and practical Implications

Blockchain technology has transformed the landscape of modern business, offering secure, transparent, and decentralized solutions to a countless of industries. Among the diverse array of Blockchain platforms, Hyperledger Fabric stands as a leading candidate, offering an enterprise-grade framework [3, 4, 5] that addresses the multifaceted demands of various applications, particularly within the financial sector. The significance of Blockchain technology lies in its promise of trustless, peer-to-peer transactions that eliminate intermediaries, reducing cost and increasing security. Hyperledger Fabric has emerged as a robust solution for enterprises seeking to harness the potential of Blockchain. It brings with it a range of features that enable the creation of private, permissioned networks, ensuring confidentiality and scalability in a world where data privacy and security are dominant [6, 7, 8]. The motivation behind this work is to inspect the Hyperledger Fabric's performance under various operational conditions and network scenarios since performance is pivotal to the success and scalability of any Blockchain-based system. In situations where financial transactions, supply chain management, and even healthcare rely on the Blockchain's immutability, the need for understanding the performance dynamics of Hyperledger Fabric has become the basic need of today's era.

1.3 Benchmarking Tool used: Hyperledger Caliper

Hyperledger Caliper is a Blockchain performance benchmarking tool which facilitates performance testing of different Blockchain platforms under varying load conditions. It is specifically designed for assessing the performance of Blockchain platforms, including Hyperledger Fabric, Hyperledger Sawtooth, and others. These parameter details are helpful for optimizing existing Hyperledger Fabric Blockchain solution and are crucial for the design of new, more efficient, and scalable Hyperledger Fabric based applications.

The remainder of this paper is organized as follows. Section 2 details the methodology used for performance testing. In section 3, we have discussed a theoretical model used for validation of results. Results and their implications are discussed in Section 4. Finally, Section 5 concludes the paper and brief about future work.

1.4 Major Contributions

The major contributions of the research article on the performance analysis and use of mathematical model

[15] to validate results of Hyperledger Fabric can be summarized as follows:

1. **Comprehensive Performance Evaluation:** The article provides comprehensive performance analysis of Hyperledger Fabric Blockchain platform. It offers important insights about the platform's behaviour under various workloads and network conditions, addressing challenging aspect of Blockchain technology adoption.
2. **Theoretical Analysis with Queuing Theory based Mathematical model:** To validate obtained simulation results, queuing theory based mathematical model have been used and detailed analysis of results is prepared. This comparison provides a structured framework to understand and predict the behaviour of Hyperledger Fabric in terms of transaction processing and resource allocation.
3. **Scalability Insights:** Through experiential data and mathematical analysis, the article offers insights into the scalability limits of Hyperledger Fabric. It sheds light on how the platform performs as the network grows, aiding in decision-making for organizations eyeing to scale their Blockchain applications.
4. **Cross-Industry Relevance:** The research findings can be useful in domains beyond finance, extending to supply chain management, healthcare, and other sectors where Hyperledger Fabric is being employed.

2. Methodology

2.1 Integrating Hyperledger Fabric with Hyperledger Caliper

Hyperledger caliper tool requires three input files namely network configuration file, benchmark configuration file, workload module along with benchmark artifacts shown in Fig. 1.

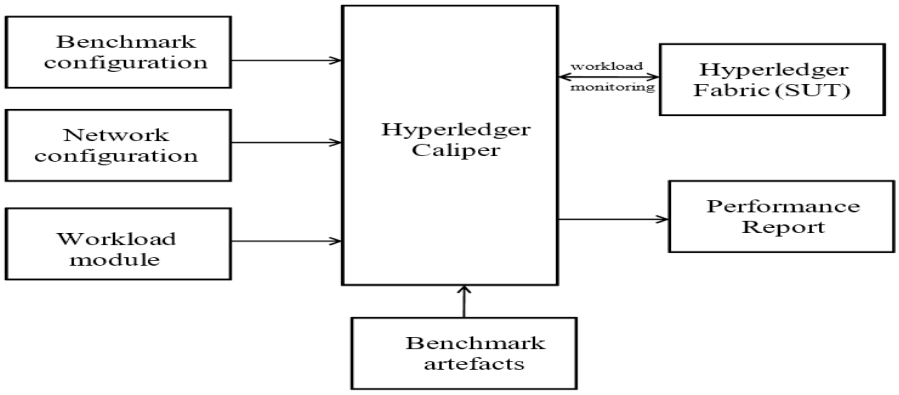


Fig.1. Hyperledger Caliper Architecture

To carry out the performance analysis, we have configured the benchmark configuration for varying number of transactions, transaction arrival rates and block sizes.

2.2 System setup and configuration:

To evaluate the performance of Hyperledger Fabric, an instance of a Hyperledger Fabric platform is deployed on AWS (Amazon Web Console) cloud platform. The deployment model is as shown in Fig. 2. All experiments are conducted on ubuntu18.04 (64 bit) running in environment [Intel(R) Core™ i5-1135G7 @ up to 4.2 GHz processor] with 8 GB RAM.

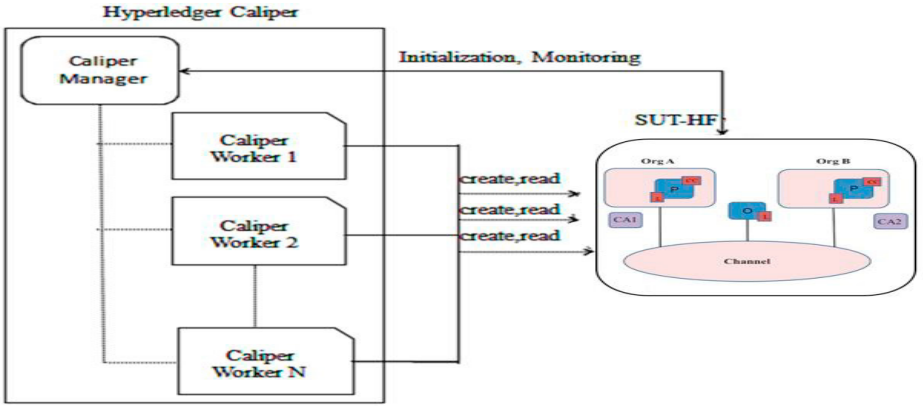


Fig.2. System Setup for Performance Analysis

We have developed Hyperledger Fabric test framework with 2 and 4 organizations with 1 peer per organization, solo ordering service, Certificate Authorities (CA) and single channel. Developed HF network (With CBFT chaincode logic) is a System Under Test (SUT) to be evaluated for performance as shown in figure 2. Each component of HF network is launched as a separate Docker Container. All Docker containers are running on a single physical machine. In Hyperledger Fabric, peers representing each organization handles transaction execution and maintains the ledger, orderer is responsible for ordering and packing the transactions in a block [9, 10, 11, 12] and broadcasting the block in the network. CA provides certificates to the participants for authentication inside the network [13, 14]. State databases supported by HF are LevelDB and CouchDB, we are using LevelDB for this deployment. Smart contract is written in JavaScript and Node.js client Software Development Kit (SDK) is used to interact with HF network.

3. Theoretical Analysis

In this section, we are explaining the theoretical model engaged in our analysis, which provides the foundational framework for our examination. Subsequently, we proceed to offer an involved calculation of latency pertaining to a single-channel Fabric network. This calculation is methodically aligned with the transaction flow of execute-order-validate, explaining the various stages and complexities involved in the latency process. This analysis aims to provide a clear and insightful understanding of latency within the Hyperledger Fabric Blockchain ecosystem, underlining its significance in the dominion of distributed ledger technology.

3.1 Theoretical Model

In our performance study, we focus on a Fabric Blockchain network employing the Solo consensus mechanism. Based on mathematical model proposed by Xu X. et al. [15], we have adapted the equations for our specific use case. For this model, a single transaction processing server have been considered and it assumes transactions arrival and processing at the server at a certain rate.

The network, in turn, is composed of C distinct organizations, with each organization maintaining P peers. We have assumed that, the arrival of new transaction follows a Poisson Point Process, characterized by a rate of arrival denoted as R_{arr}. Every transaction exhibit S_p bits of packet size, while the size of an individual signature is specified as S_{sig} bits. To ensure consensus, each transaction must undergo endorsement by P_e endorsers, as per the defined endorsement policy. The packet size of a single endorsement is represented by '*a*' bit. Furthermore, our analysis considers the Blockchain network to possess a bandwidth of B bits per second (bps), coupled with an average propagation delay marked as t_{pd}. For this model it is also assumed by authors [15], that all the network nodes have essentially infinite waiting rooms, facilitating the accommodation of incoming transactions without congestion. Additionally, the nodes within the network are configured to serve transactions in a First-In-First-Out (FIFO) manner. The service time at each node adheres to an exponential distribution with a mean duration of $1/\mu$. Moreover, we have included the Table 1 for reference and clarity, which provides a brief about the notations used in

this paper. This comprehensive set of parameters and notations serves as the foundation for in-depth analysis of latency [15] within the Fabric Blockchain system.

Table 1: Model Parameters

Parameter	Description
R_{arr}	Arrival rate of transactions
K	Organization
C	Client
P	Peers per organization
X	Total channels in the network
P_e	Endorser peers
S_p	Transaction proposal packet size
a	Size in bits of a single endorsement
S_{sig}	Signature size
μ	Service rate
S_B	Size of a block
t_B	Block interval
B	Network bandwidth
t_{pd}	Delay involved in propagation
P_A & P_B	Probability of block propagation for two different scenarios
Y_e	Traffic intensity at an endorser node
L	Total Latency
T_w& T_b	Waiting time and batch time

3.2 Latency analysis for single-channel solo-orderer Hyperledger Fabric

Here we are discussing latency performance modelling of Hyperledger Fabric Blockchain network. Total transaction latency (L) has been considered as combination of execution phase latency (L_e), ordering phase latency (L_o) and validation phase latency (L_v).

Total Transaction Processing Latency = Latency (Execution phase + Ordering phase + Validation phase)

$$L=L_e+L_o+L_v \quad [1]$$

3.2.1 Execution phase latency (L_e):

In the context of a Hyperledger Fabric Blockchain network, the transaction proposal process is characterized by certain parameters. Clients initiate this process by sending transaction proposals, each consisting of 'S_p' bits of data. The transferring latency associated with these transaction proposals plays a significant role in the network's overall performance which is calculated as:

$$\text{Transferring latency} = S_p/B + t_{pd} \quad [2]$$

To assess the traffic intensity at an endorser node (Y_e), we have utilized following formula:

$$Y_e = R_{arr}\, S_p/\mu \quad [3]$$

Furthermore, to determine the processing latency at each endorser node (L_p), Little's Law is used.

$$L_p = 1/\mu\,(1 - Y_e) \quad [4]$$

As the transaction proposal phase progresses, the endorser process these proposals and subsequently sending these endorsements to the client. The endorsements, each comprising 'a' bit of data, are transmitted to the client. The transferring latency for these endorsements is calculated as $a/B + t_{pd}$, taking into account the bandwidth (B) and the propagation delay (t_{pd}) involved in the data transfer. The collective latency for the entire execution phase (L_e) in the network encompasses all these components and their associated latencies.

The overall latency for the execution phase L_e is

$$L_e = (S_p + a)/B + L_p + 2\, t_{pd} \quad [5]$$

3.2.2 Ordering phase latency (L_o):

In the process of transitioning from the endorsement phase to the ordering phase in a Hyperledger Fabric Blockchain network, several key steps are involved:

- **Client Transaction Assembly and Transfer:** After successfully collecting the required endorsements for a transaction, the client forms new proposals by including these endorsements. The resulting proposals include transaction proposal (S_p), P_e endorsements each with size 'a', and one client signature (S_{sig}). The client then transmits it to the orderer, a critical step in consensus of the network. This transferring latency is calculated based on time required by the client to send the transaction to the ordering service, denoted as

$$(S_p + P_e\, a + S_{sig})\ /B + t_{pd} \quad [6]$$

- **Ordering Service Block Generation:** The ordering service is responsible for new block generation with all incomplete transactions received from different clients. After receiving these transactions, orderer organizes and sequences them into a predefined order, ready for inclusion in a block. The orderer groups these transactions together in blocks using a pluggable consensus mechanism, which ensures that transactions are ordered and recorded consistently across all nodes in the network. We are using solo ordering service. Latency of ordering phase is given by:

$$L_{orderer} = L_{wait} + L_{batch}. \quad [7]$$

In solo ordering, block get added in the Blockchain only when it satisfies the one of the following criteria:

1. Configured block interval T_B is passed from the time at which the last block is added to the chain.
2. When Block is full that is number of pending transactions is equal to the configured Block Size S_B.

Probability for first condition P_A is calculated as

$$P_A = \sum_{i=0}^{S_B} [(R_{arr}\, t_B)^i / i!] * e^{-(R_{arr} * t_B)} \quad [8]$$

And

$$P2 = 1 - P1 \quad [9]$$

$$T_{total} = \sum_{j=0}^{S_B - 1} \sum_{i=j}^{S_B - 1} T_i \quad [10]$$

Finally, value of T_w and T_b in equation (9) is calculated as:

$$L_{wait} = P_A\, t_1 + P_B\, t_2. \quad [11]$$

$$L_{batch} = 1/\mu - R_{arr}\,(S_p + P_e\, a + S_{sig}). \quad [12]$$

Where, $t_1 = t_B/2$ and $t_2 = T_{total}/S_B$.

3.2.3 Validation phase latency (L_v):

In a Hyperledger Fabric Blockchain network, the validation phase is a critical step where all peers validate each transaction in a block. This process involves various components, including the verification of transactions against the Versioned Smart Contract (VSCC) and Multi-Version Concurrency Control (MVCC) validation. To calculate the total processing latency (L_v) in the validation phase, we considered the time required for these validation processes. It is calculated as:

$$L_v = 2\,(1/\mu - R_{arr}\,(S_p + P_e\, a + 2S_{sig})) \quad [13]$$

Default configuration considered for above calculations is as shown in table 2:

Table 2: Default Configuration

Parameter	Value
R_{arr}	Varying with total number of transactions
K	2,4
C	1
P	1
X	1
μ	Corresponding throughput value
S_B	200 ***Ref. [11]***
t_B	1 s ***Ref.[11]***
B	20 Mbps
t_{pd}	15 milliseconds

4. Results

In this section results for latency are presented for different configurations such as varying Block size (S_B), transaction arrival rate (R_{arr}), number of organizations (K) and block interval (t_B).

4.2.1 Results for varying Transaction Arrival Rate (R_{arr}) and Block Size (S_B):

Here we have utilized linear rate control mechanism of Hyperledger Caliper for rate control with startingTPS and finishingTPS of 100 to 500 transaction per second respectively. It increases transaction arrival rate (R_{arr}) automatically with increasing number of incoming transactions.We have repeated the experiments by varying total number of transaction (in turn transaction arrival rates) and block size which shows that latency increases with incresed block size; as larger blocks take longer to propagate across the network, more time for the consensus algorithm to agree on and more time for peers to validate more number of transactions they contain, potentially increasing the time required for a transaction to get added to the ledger hence increased resulting latecy. Also its been observed that once number of transactions exceeds 20000, latency is decreasing because more transactions are getting included in the block reducing the overhead per tansaction.

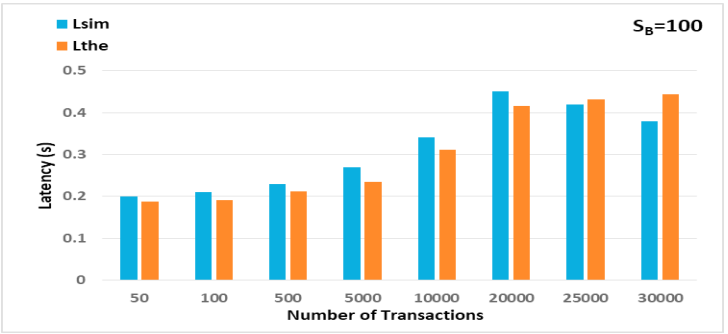


Fig.3(a). Latency vs. number of transactions with block size S_B=100 & block interval time T=1s

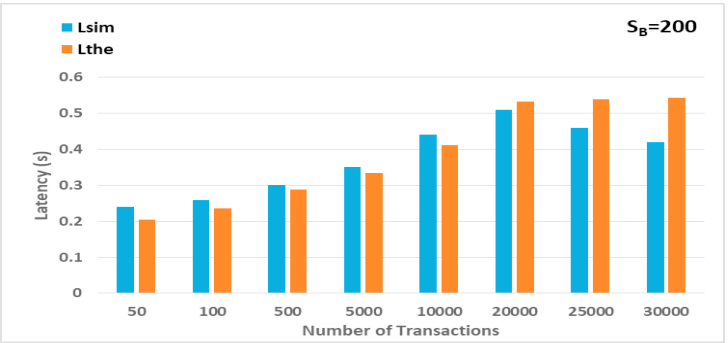


Fig.3(b). Latency vs. number of transactions with block size S_B=200 block interval time T=1s

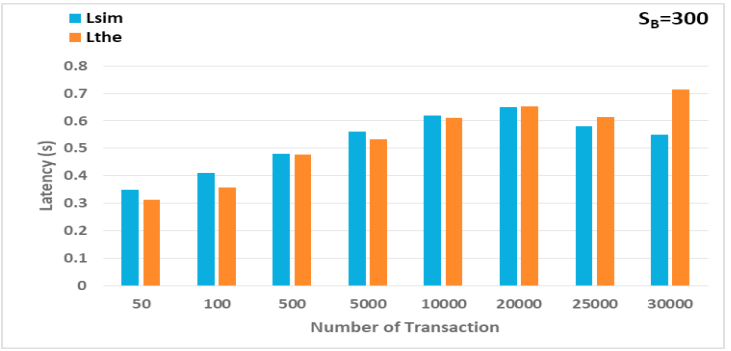


Fig.3(c). Latency vs. number of transactions with block size S_B=300 block interval time T=1s

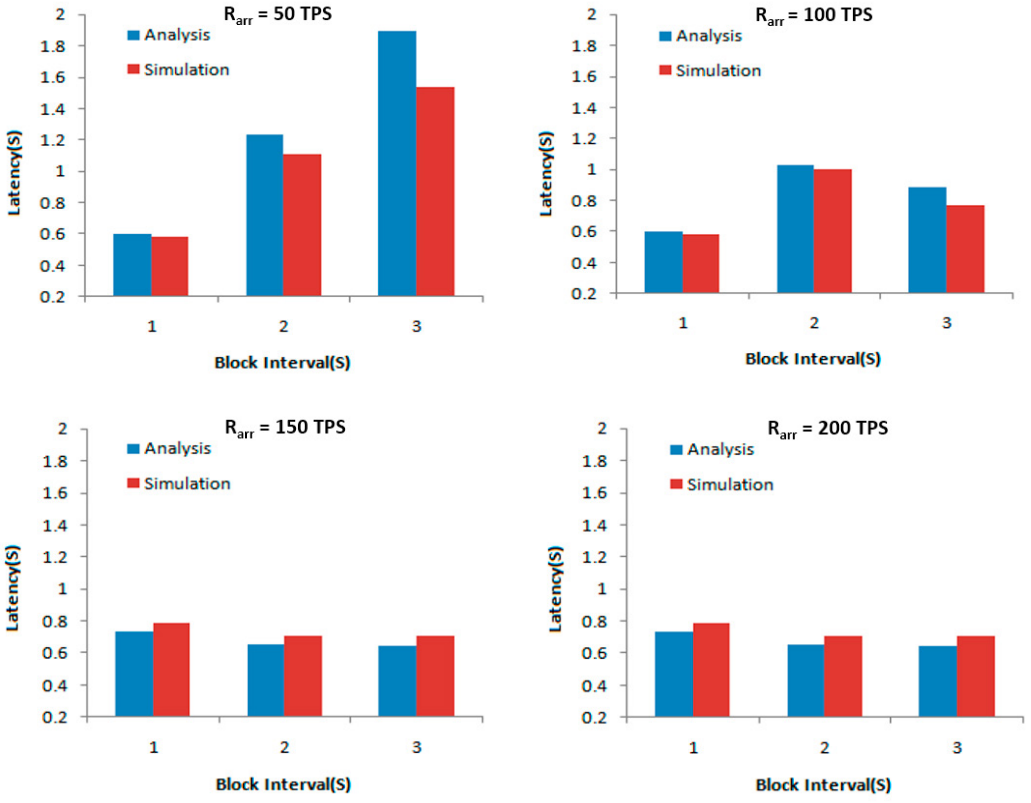
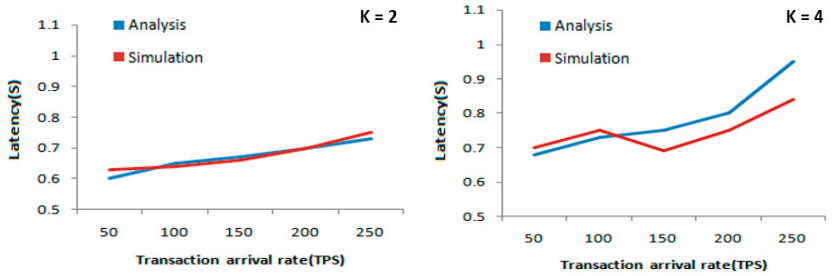


Fig.4. Latency results of HF network with varying bock intervals (1s, 2s, 3s) with constant block size S_B=200

Further, we have analysed effect of varying Block Interval Time *(t_B)* at different transaction arrival rates as shown in figure 4. To generate figure 4, we have varied block interval time by changing BatchTimeout parameter value in configtx.yaml file from 1 to 3. Again, transaction arrival rate is varied by utilizing rate control mechanism of Hyperledger Caliper as mentioned above. From obtained results, it is clear that, latency is minimum at lower transaction arrival rates and lowest block interval that is 1s. A longer block interval time increases the time required for a transaction to get added to the Blockchain. As transactions are typically included in blocks, and if blocks are generated infrequently, transactions will spend more time waiting to be confirmed. But it's been observed that at

higher transaction arrival rates, latency is almost independent of block interval because block size limit is getting reached faster at higher transaction arrival rates.

4.2.3 Results for varying organization numbers:



No. of organizations = 2 **No. of organizations = 4**

Fig.5. Latency when number of organizations=2, 4

Latency tends to increase with number of organizations as shown in figure 5 above because of several factors such as consensus overhead, communication overhead and Endorsement policy requirement. More organization implies that all participating organizations agree on the order and validity of transactions. So, with the increasing number of organizations consensus process becomes more complex and time-consuming as each organization needs to reach consensus. Also, with more organizations, there is increased communication overhead between peers, endorsers, and the orderer nodes causing delay in the process. Further AND endorsement policy used here takes more time to execute, hence introducing additional latency.

4.2.4 Simulation results and validation with theoretical model

Model accuracy under varying numbers of transactions and Block Size is shown in table 3.

Table 3(a, b, c): Model accuracy under varying number of transactions and Block Size

No. of Transactions	L_{sim}	L_{the}
50	0.2	0.187
100	0.21	0.191
500	0.23	0.212
5000	0.27	0.234
10000	0.34	0.311
20000	0.45	0.416
25000	0.42	0.432

S_B=100

No. of Transactions	L_{sim}	L_{the}
50	0.24	0.205
100	0.26	0.236
500	0.3	0.289
5000	0.35	0.334
10000	0.44	0.412
20000	0.51	0.532
25000	0.46	0.539

S_B=200

No. of Transactions	L_{sim}	L_{the}
50	0.35	0.312
100	0.41	0.356
500	0.48	0.476
5000	0.56	0.532
10000	0.62	0.612
20000	0.65	0.654
25000	0.58	0.614

S_B=300

5. Conclusion and Future Scope

In this research, we conducted a comprehensive performance modelling and analysis of a Hyperledger Fabric based Cross-Border Fund Transfer Blockchain Network. Our study aimed to evaluate the system's efficiency, scalability, and reliability to ascertain its viability for real-world cross-border financial transactions. The key findings and insights from our research can be summarized as follows:

1. **Latency Optimization:** From our analysis it is clear that careful configuration (block size, Block interval) and optimization of the Hyperledger Fabric network (Transaction arrival rate, endorsement policy and consensus mechanism used) can considerably contribute to reduced transaction latency.
2. **Endorsement Policies:** The choice of endorsement policies plays crucial role in governing the validation process for transactions.
3. **Real-World Applicability:** It is confirmed that Hyperledger Fabric has the potential to serve as a foundation for cross-border fund transfer solutions, as long as that network administrators carefully design, configure, and manage the system to meet the specific demands of the given use-case.

In conclusion, our study has provided potential insights into the performance characteristics of a Hyperledger Fabric based Cross-Border Fund Transfer Blockchain Network. Further, implementing load balancing and maintaining continuous monitoring to take appropriate actions for performance improvements are measures to reduce latency in a Hyperledger Fabric-based Cross-Border Fund Transfer Blockchain network; these methods represent our ongoing and future work. Along with performance, reliability of Blockchain network can be improved by designing the network with scalability in mind to accommodate increasing transaction volumes. To improve scalability one can choose hardware scaling, dynamic membership management or advanced network topologies. The results of this research can inform the development and deployment of secure, efficient, and scalable Blockchain solutions for the financial sector. As Blockchain technology continues to mature, we anticipate its adoption for cross-border fund transfer services, contributing to enhanced security, transparency, and efficiency in the global financial landscape. Our future work may involve further research and practical implementations with different network configurations to improve and validate our findings in real-world scenarios.

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